

Multi-Modal Sensors Fusion for Fall Detection and Action Recognition in Indoor Environment

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Abstract— In computer vision and inertial sensors, recent research focuses on monitoring and identifying falls, as well as daily actions, through autonomous human action detection. In this paper, the proposed model utilizes multi-modal sensors for fall detection. The data from inertial sensors was filtered using a bilateral filter, known for its monotonic and maximum flat magnitude response in the passband, giving smoothness. Key properties, such as the Gaussian Mixture Model (GMM) and parseval's energy, were retrieved. Additionally, fall detection was accomplished utilizing RGB (red-green-blue) videos, where features such as rectangles, triangles, full-body ridges, and full-body curves were generated. The final results from the inertial sensor and vision data were fused utilizing multimodal fusion. We then optimized the fused data using the Naive Bayes approach and trained Multi-layer perceptron (MLP) classifier for classification. The UR Fall Detection (URFD) dataset was utilized to evaluate the proposed system, demonstrating its usefulness by reaching an accuracy of 88%.

Keywords— Sensors processing, Human fall detection, Human action recognition, Human computer interaction, Naïve Bayes, Applications of machine learning.

I. INTRODUCTION

Over the last few years, human fall detection has made considerable breakthroughs, earning increased interest in numerous industries. Many researchers have incorporated depth, vision, and inertial sensors [1], to boost the performance of their system. Human action recognition has proven effective in multiple sectors, including security monitoring, human-computer interaction (HCI) [2-4], human health care [5-8], assistive technology [9-10], and gait abnormality detection [11-14]. We have employed both vision data and inertial data to produce better outcomes. The procedure of human fall detection generally involves data preprocessing [15-21], feature extraction [22-26], feature fusion [27-31], feature optimization [32], and feature classification [33-34]. However, getting reliable recognition during fall detection is tough due to the restrictions of using inertial, vision, and depth sensors independently.

Human fall detection utilizing vision-based and inertial-based sensors has its limits [35-37]. Vision-sensors pose issues for numerous reasons, such as inadequate illumination, variable body motions, and varied views. Besides, inertial sensors come with intrinsic limitations, including signal and ambient noise, challenges in computing, and analysis induced by changeable placement. Moreover, wearing one or multiple sensors at once can create pain and undermine security for the user.

II. LITERATURE REVIEW

Several scholars have offered ways to promote society well-being, particularly in human action recognition (HAR) and

fall detection. [38] developed an elliptical modeling-supported system using RGB drone videos, incorporating Felzenszwalb's segmentation and a Gaussian mixture model for improved HAR from aerial views. [39] proposed a system incorporating videos and inertial sensors for fall detection by logistic regression. [40] utilized wearable sensors, GPS, CNN, and LSTM for outdoor HAR, obtaining high localization and motion identification. [41] designed a fall detection system for long-term care, incorporating accelerometers, RGB, and depth sensors, reaching using CNN and GRU models. Similarly, [42] proposed an object detection system using Maximum Entropy scaled Super-Pixels and deep belief networks, excelling in dense situations with minimal processing costs. Despite these gains, obstacles exist in adjusting to different settings. Therefore, we propose a novel framework by combining multi-sensors to develop sturdy multi-feature descriptors for human fall detection.

The paper is structured as follows: Section III explains the suggested framework, encompassing preprocessing, windowing, silhouette recognition, skeleton models, and feature extraction, fusion, optimization, and classification. Section IV offers experimental data and comparisons with various fall detection systems. Section V finishes the paper.

III. SYSTEM DESIGN

We tested the effectiveness of the proposed method using the URFD dataset., incorporating both vision and inertial sensors for fall recognition. vision sensor data processing contains four stages: preprocessing, human silhouette detection, skeleton modeling, and feature extraction; geometric shapes, ridges, and curves. Inertial sensor data processing similarly follows four stages: preprocessing, signal windowing, feature extraction such as Parseval's energy or GMM, and feature fusion. Classification is done using the help of the MLP classifier. Visualization of the system architecture is presented in fig. 1.

1. Data Preprocessing of inertial sensor

Inertial sensing data preprocessing is done by first applying a kalman filter to the signal in order to mitigate noise. It decreases the mean square error (MSR) successfully and calculates the current signal from earlier data for proper filtration. The kalman filter equations is shown below.

$$s_{n|n} = s_{n|n-1} + G_n(m_n - C s_{n|n-1}) \quad (1)$$

Where $s_{n|n}$ is the revised state estimate, accounting for the additional measurement. And $s_{n|n-1}$ is the predicted state estimate. G_n is the kalman gain. m_n is the current measurement (in this instance, the acceleration magnitude) C defines the measurement matrix. Fig. 2 shows the results of the kalman filtered signal.

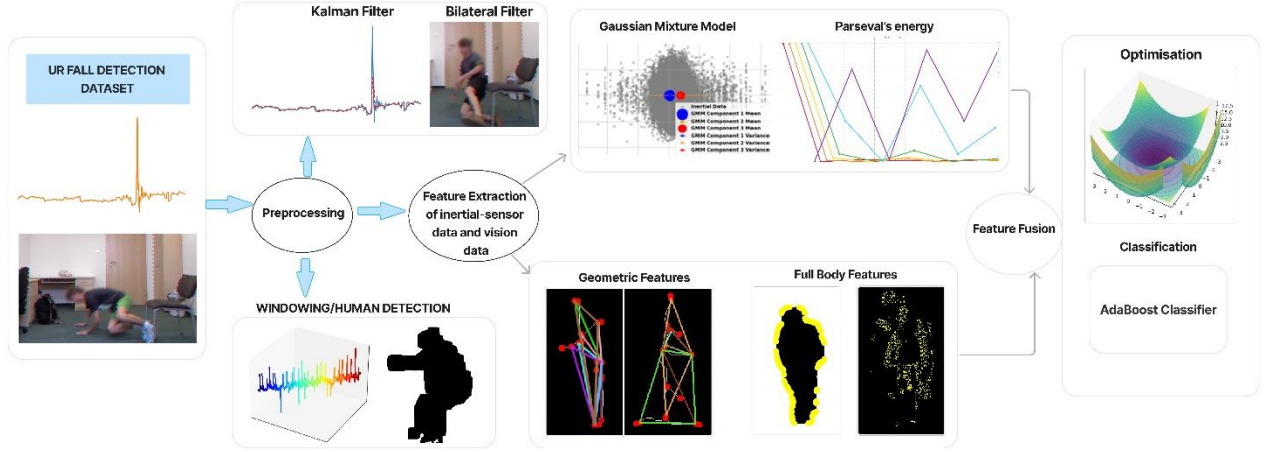


Fig. 1. The Architecture of Proposed system for human fall detection and action recognition system.

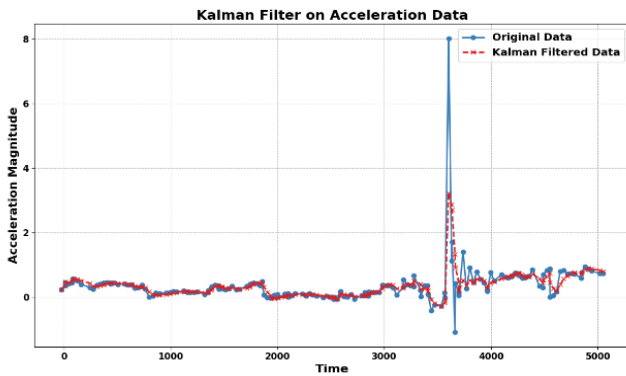


Fig. 2. Raw inertial-sensors versus filtered inertial-sensors data over the URFD dataset.

2. Preprocessing of vision-sensor data

The vision-sensor data was preprocessed using a bilateral filter [43] to help reduce extraneous information from frames. The equation of the bilateral filter is provided below.

$$J_p(a) = \frac{1}{N_a} \sum_{b \in s} F_\sigma(|a - b|) F_\sigma(J(a) - J(b)) J(b) \quad (2)$$

Where, $J_p(a)$ represents the filtered intensity at pixel a . N_a is the normalizing factor ensuring the total weights equal 1. s denotes the spatial neighborhood around pixel a . $F_\sigma(|a - b|)$ is a Gaussian function based on the spatial distance between pixels, reducing the influence of distant pixels. $F_\sigma(J(a) - J(b))$ is a Gaussian function based on intensity differences between pixels a and b [44]. The frames were then converted to grayscale to reduce noise, as represented in fig. 3.

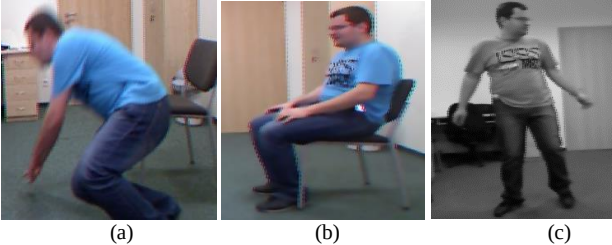


Fig. 3. RGB images preprocessing (a) shows RGB image (b) shows filtered image and (c) shows grayscale image over URFD dataset.

A. Windowing and Human Silhouette Detection

In this section, signal windowing and human silhouette detection is briefly described below.

1. Inertial-sensors windowing

Initially, the signals were filtered using the kalman filter. after that, the filtered signals were concatenated and then split using a sliding window technique [45-47] with a 10-second window size, allowing effective analysis of static posture and dynamic motion. The signal windowing equation is provided below.

$$v(m) = P_1 + P_{m+1} + \dots, P_{m+m_{len}} - 1 \quad (3)$$

Where, $v(m)$ represents the m -th windowed addition of the signal and P denotes the signal. w_{len} represents the size of the window. Fig 4 represents the results of signal windowing.

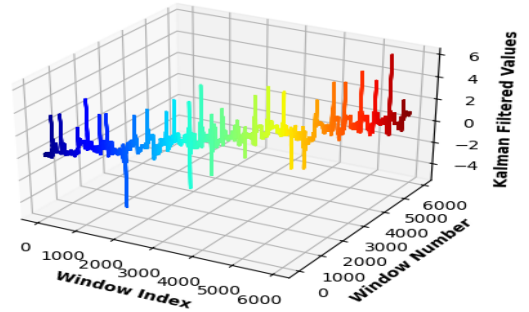


Fig. 4. The results of windowing on inertial-signal data over URFD dataset.

2. Human Silhouette Detection

For visual features, a bilateral filter was applied and grayscale images used to enhance segmentation quality and eliminate noise for the objective of detecting human silhouettes [48-53]. In the process of human silhouette detection, back ground subtraction technique is used to separate the object of interest from the images. The following is the equation that was used to do background subtraction.

$$Q(Z) = \sum_{c=1}^M u_c \cdot \phi(A; v_c, \Omega_c) \quad (4)$$

Where, $W(Z)$ represents the likelihood of observing pixel value (Z). M shows the number of gaussian distributions. u_c is the weight of the c -th gaussian component. While, $\phi(A; v_c, \Omega_c)$ shows a gaussian probability density function with mean v_c and covariance Ω_c . After that, adaptive thresholding was employed to generate a binary mask. The equation of adaptive thresholding is provided below.

$$S(p, q) = \frac{1}{V} \sum_{(a,b) \in V} J(a, b) - D \quad (5)$$

Where, $S(p, q)$ is the threshold value for pixel (p, q) and $J(a, b)$ is the intensity of pixel (a, b) in the window and D is a constant. Next, contour detection was done for identification of important forms, and mask refining by focusing on the largest contour. The equation for determining the greatest contour is given below.

$$F = \sum_{(a,b) \in Q} J(a, b) \quad (6)$$

Where Q denotes the contour. Lastly, mask refinement focuses on the contour. The silhouette is extracted by inverting the mask:

$$\text{shape}(a, b) = 255 - \text{mask}(a, b) \quad (7)$$

This swaps the foreground pixel value to the background pixel value (white with black), while the background pixel value is switched with the foreground value (black with white), thereby creating silhouette detection. Fig. 5 represents the results of human silhouette detection.

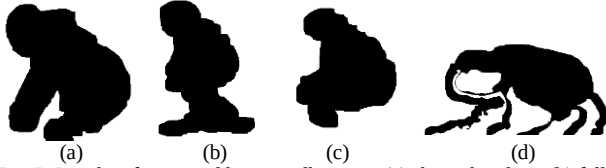


Fig. 5. Results of extracted human silhouettes (a) shows bending, (b) falling, (c) standing, and (d) walking over the URFD dataset.

B. Skeleton model

In this section, the human skeleton [54-55] is explored in the context of human fall detection and action recognition. The skeleton efficiently reflects the body, enabling easy identification of specific body parts and extraction of movement data for action classification. We employed the skeletonization method in conjunction with Otsu's thresholding methodology to extract skeletons. First, grayscale photos are adjusted to a pixel range of (0,1). The photos are subsequently flipped to emphasize the object (white) against a black background for thresholding purposes. Otsu's approach is used to automatically identify the threshold separating the human silhouette from the background. The equation for Otsu's thresholding is provided below.

$$\sigma_c^2(t) = p_1(t) \cdot p_2(t) \cdot [\mu_1(t) - \mu_2(t)]^2 \quad (8)$$

Where, $\sigma_c^2(t)$ represents the between-class variance, $p_1(t)$ and $p_2(t)$ denote the probabilities of two classes separated by threshold t . $\mu_1(t)$ and $\mu_2(t)$ indicate the class means. Visualization of skeleton model is given in fig. 6.

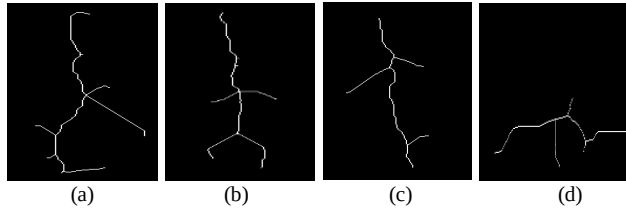


Fig. 7. Results of skeleton model (a) shows walking (b) and (c) represent standing, and (d) represents falling, over the URFD dataset.

C. Feature Extraction

Feature extraction is vital for proper classification and medical monitoring. Parseval's energy and GMM from inertial-sensors, combined with full-body ridges, curves, and geometric shapes from vision sensors, are extracted to increase system performance and reduce misclassification.

1. Feature extraction of inertial-sensor data

Initially, we estimated Parseval's energy for windowed data to quantify signal power, revealing insights into action patterns. Using the sum of squares formula, we computed and normalized energy quantities, then allocated actions. We provide Parseval's energy equation below.

$$E = \sum_{i=1}^M y[i]^2 \quad (9)$$

Where, E represents the computed parseval's energy. M denotes the total number of data points within the window. whereas $y[i]$ represents the i -th data point within a window. Illustration of parseval's energy is given in fig. 7

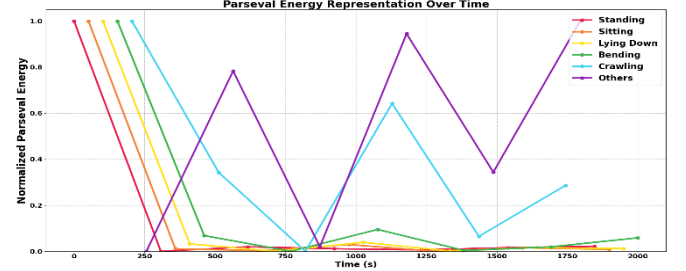


Fig. 8. Results of Parseval's energy of various actions over the URFD.

After that, GMM was applied for density estimation and clustering, calculating the mean (μ), covariance (Σ), and weights (π). We have utilized an iterative maximizing technique to discover which weight vector has the highest likelihood. Afterward, the GMM covariance is calculated to measure the difference between two samples within the current signals. The following equation has been applied to obtain GMM.

$$P(x) = \sum_{k=1}^K \pi_k \cdot \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma_k|^{\frac{1}{2}}} \exp\left(-\frac{1}{2} (x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k)\right) \quad (10)$$

Where, π_k is the a maximum weight for the k^{th} gaussian component, μ_k is the mean vector. Σ_k is the covariance matrix, and k is the total number of gaussian components, d is the dimensionality of the data. Fig. 8 shows GMM results.

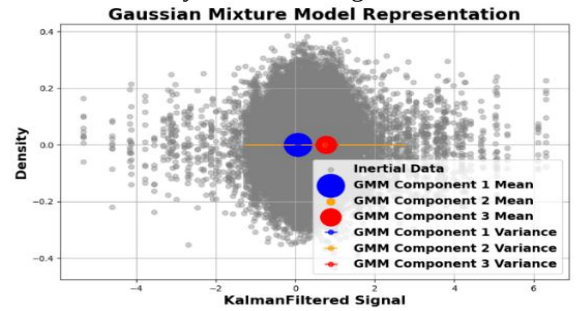


Fig. 9. GMM of windowed inertial-sensors data over the URFD dataset.

2. Feature extraction of vision-sensor data

For vision data, geometric characteristics including rectangles, triangles using keypoints, full-body ridges, and curves were computed. Triangles were created out of nine chosen keypoints, with each triangle created by three points. The angle between two vectors, formed by these points, was calculated using the dot product and vector magnitudes.

$$\theta = \arccos \frac{u \cdot v}{||u|| ||v||} \times \frac{180}{\pi} \quad (11)$$

Where u denotes the vector from point 2 to point 1. v shows the vector from point 2 to point 3. While, $(\cdot)$ denotes the dot product of two vectors. While $||u||$ and $||v||$ indicates

magnitudes (Euclidean norms) of vectors u and v . θ is angle between the two vectors. Visualization of triangles using keypoints is given in fig. 9.

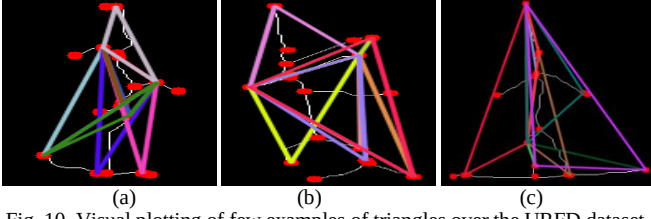


Fig. 10. Visual plotting of few examples of triangles over the URFD dataset.

As, Rectangles were recreated by joining four keypoints in a frame. This strategy is depicted in fig. 10 where the keypoints are joined to make the rectangles.

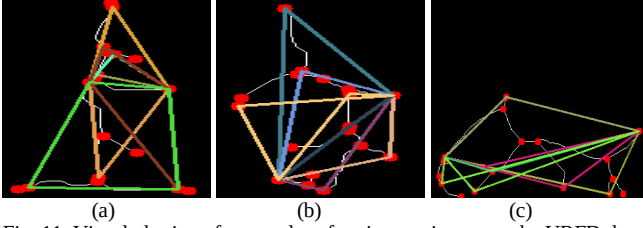


Fig. 11. Visual plotting of rectangles of various actions over the URFD dataset.

As, we used the full-body ridge feature to highlight human body contours and key anatomical features. A hessian matrix and its eigenvalues were applied to capture structural aspects of body posture for action recognition. The formula for calculating the eigenvalues is provided below:

$$u_{1,2} = 0.5(D_{xx} + D_{yy} \pm \sqrt{(D_{xx} - D_{yy})^2 + 4D_{xy}^2}) \quad (12)$$

In this context, D_{xx} and D_{yy} represent second-order derivatives of image intensity along x and y directions, respectively, while D_{xy} represents mixed second-order derivative. Ridges are extracted where both eigenvalues u_1 and u_2 are negative. Fig. 11 shows the results of the ridges.

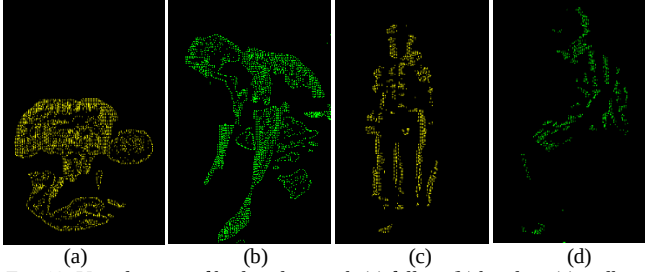


Fig. 12. Visualisation of body ridges with (a) falling (b) bending (c) walking (d) sitting over the URFD dataset

As, the body curve has been recognized utilizing freeman chain coding and canny edge detection for identifying critical areas on the contour. Freeman chain coding leverages a priori determined directed vectors to encode the edge of an object. Directions are often described as the following:

$$V = [(1,0), (1,1), (0,1), (-1,1), (-1,0), (-1,-1), (0,-1), (1,-1)] \quad (13)$$

Similarly, all the descriptors of silhouette have been computed. The results of the body curve are shown in fig. 12.

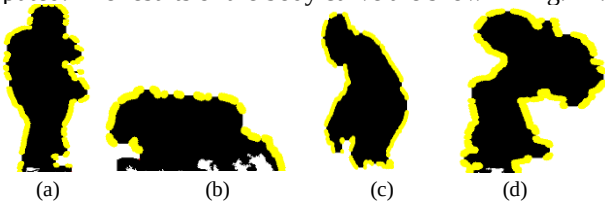


Fig.12 Results of body curve with (a) standing (b) falling (c) bending (d) sitting over the URFD dataset.

D. Feature Fusion

In feature fusion, inertial-sensor data and vision data are analyzed individually. Inertial features (e.g., Parseval's energy, GMM) and vision features (e.g., geometric features, full-body ridge, curve) are concatenated, then fused based on a common column. The fusion equation is illustrated below.

$$z_{fused}(t) = [z_1(t)||z_2(t)] \quad (14)$$

Where, $z_{fused}(t)$ represents the fused output, $z_1(t)$ and $z_2(t)$ denote two different data modals being concatenated. Finally, fig. 13 represents fused multimodal data.

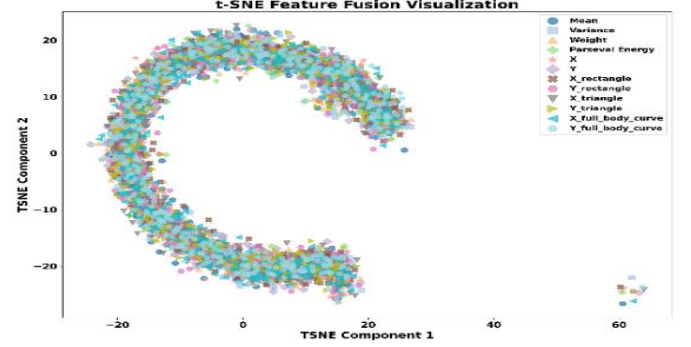


Fig. 13. Feature fusion results over the URFD dataset.

E. Feature Optimisation

The Naïve Bayes Optimizer has been used in this paper for feature optimization because of its computational efficiency, which is especially beneficial for real-time fall detection systems. Naïve Bayes offers competitive accuracy at a lower computational cost than more sophisticated optimization methods like support vector machines or metaheuristic algorithms. It efficiently selects the relevant number of features by calculating the conditional probability of each feature based on a class label. What makes it even more appropriate for such constraints is that wearable technology requires high levels of accuracy and depth as well as speed. The probability of a class based on a certain feature is computed using the following equation:

$$P(C_i|X_j) = P(C_i) \times P(F_j|C_i) \quad (15)$$

Where, $P(C_i|X_j)$ is the probability of class C_i given data point F_j . $P(C_i)$ represents the probability of class C_i , and $P(F_j|C_i)$ is the conditional probability for feature F_j given class (C_i) . In feature optimization, features are evaluated for class probability, and the top k with the highest mutual information and correlation are selected. Fig. 14 shows the results on the UR-Fall detection dataset.

Naive Bayes over URFD dataset

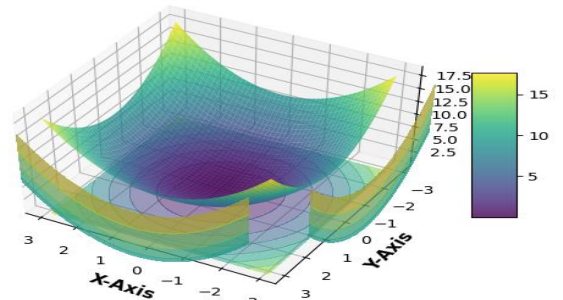


Fig. 14. Feature optimisation results over the URFD dataset.

F. Classification

The MLP classifier in this work comprises an input layer, hidden layers, and an output layer. The dataset was split 4:1 into training and testing sets. To reduce the class imbalance problem, synthetic minority over-sampling technique (SMOTE) was used to generate synthetic samples of the minority class. Using StandardScaler, we preprocessed the data, and in the next step, the hyperparameter tuning was performed using the GridSearchCV, and for features, we adjusted the hidden layer nodes, the activation function (tanh), learning rate, and regularization. This was done to avoid overfitting, and early stopping was used in the process. The classifier attained a test accuracy of 88% with regards to performance, using accuracy and classification reports. Class imbalance and optimization problems were handled effectively. The computation at each layer is represented by:

$$Z^{(k)} = g(M^k z^{(k-1)} + (c)^{(k)}) \quad (16)$$

Where, $Z^{(k)}$ denotes the output of the $k - th$ layer. $M^{(k)}$ shows the weight matrix connecting the $(k - 1) - th$ layer to the $k - th$ layer. $c^{(k)}$ denotes the bias vector for the $k - th$ layer, g is the activation function. While $z^{(0)}$ represents the input data to the network (the input layer).

IV. EXPERIMENTAL RESULTS

In this section, we discuss the data set, illustrate the outcomes of the experiment, and accuracy of the proposed model. We also evaluate the proposed model with other known models of fall detection. In the case of the experiments, an n-fold cross-validation approach was applied to divide the data into the training and test sets.

A. UR-Fall Detection Dataset

The fall detection data was obtained using two Kinect cameras connected via USB and an IMU device worn at the waist, which communicated via bluetooth. Camera 0 and the accelerometer captured daily living (ADL) actions. In an office experiment, five volunteers underwent 70 sequences, consisting of 30 falls and 40 other typical actions. Participants purposefully fell onto a 2-cm-thick carpet, with the X-IMU wrapped around their pelvis. Each participant completed lateral, backward, and forward falls at least three times. Fig. 15 depicts the actions covered in the URFD dataset.

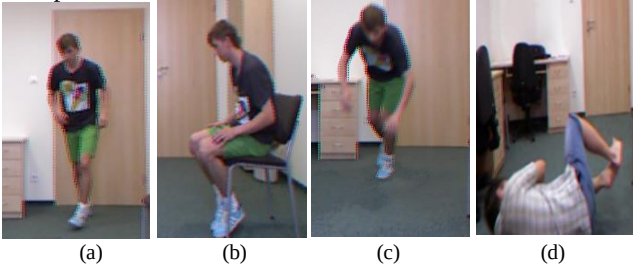


Fig. 15. Examples of various actions over the URFD dataset.

B. Experimental Results

To evaluate the classifier, a confusion matrix that represents true positives, false positives, true negatives, and false negatives was constructed. In another experiment, we used evaluation measures [56-58] comprising accuracy, precision, recall, and the F1 score, as stated in tables I and II. Furthermore, in Table III, the suggested method is compared to other current techniques in terms of fall detection accuracy on the URFD dataset.

TABLE I. CONFUSION MATRIX OVER FALL DETECTION DATASET.

URD	ST	SI	LD	BD	CR	OT
ST	86	4	3	2	2	1
SI	2	86	3	2	2	1
LD	3	3	86	2	1	2
BD	3	2	3	85	2	2
CR	3	2	2	3	85	2
OT	4	2	2	2	3	85
Mean	88%					

ST= standing; SI=sitting; LD=lying down; BD=bending; CR=crawling; OT=others.

TABLE II. PRECISION, RECALL AND F1 SCORE RESULTS OBTAINED BY PROPOSED METHOD AS MLP CLASSIFIER.

URFD dataset	Precision	Recall	F1 score
Standing	85	88	86
Sitting	87	90	88
Lying down	87	89	88
Bending	89	88	88
Crawling	89	88	88
Others	91	87	89
Mean	88	88	88

TABLE III. COMPARISON OF OUR PROPOSED METHOD ON UR FALL DETECTION DATASET WITH OTHER CUTTING-EDGE-METHODS.

Methods	Recognition Accuracy (%)
D. R. Beddiar et al., [59]	85%
Nizam, Y. et al., [60]	88.57%
Proposed approach	88%

V. CONCLUSION

In this paper, we built a multimodal fall detection system based on vision and inertial data and achieved 88% accuracy on the URFD dataset. The system contains seven phases, including data preprocessing, feature extraction, and classification by utilizing MLP classifier. visual data offers body and geometry features, while inertial-sensors provides GMM and energy features, which increase accuracy. Some of the challenges include light and noise sensitivity.

In future, we want to add deep learning classifiers for better temporal recognition, and compare the more complicated datasets, and incorporate depth and heat sensors to overcome environmental constraints including occlusion and inadequate light.

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